Tensor Algebra

Than Lwin Aung

Linearity

First of all, let us define “Linear” more abstractly, which makes a **Linear Space**.

Algebraically, **Linearity** is defined as follows.

V is a **set** for which it is possible to define two **operations**:

$$+: V \times V \rightarrow V$$
$$.: \alpha \times V \rightarrow V$$

which implies:

$$+(u, v) \rightarrow u + v$$
$$.(a, u) \rightarrow au$$

Linear Space

Any space, defined by **Linearity**, is called **Linear Space**, or **Vector Space**.

Now, what is really interesting about Linear Space is it defines what can exist in that space or not.

In other words, it defines what can be put together and what cannot be.

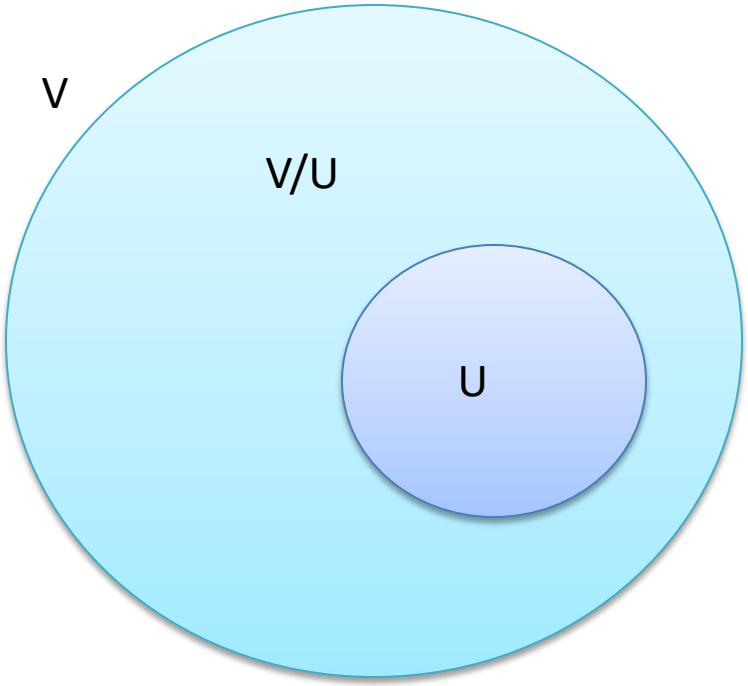
For example, the addition of two apples will always be apples; however, the addition of an apple or orange will not.

It is because the addition of an apple and an orange is not linear.

It is highly important because later we might be combining different entities together, such as Text and Image.

Subspace and Quotient Space

Any subset **U** of **V** is called Linear **Sub Space** of **U**, which is also a Linear Space. And the **Quotient Space** is defined by **V / U**, which is a **disjoint** sub space which is outside of **U** but inside of **V**.



Linear Independence

From the perspective of Subspace, we can define **Linear Independence**.

Let **X**, **Y**, **Z** $\subseteq$ **V** as disjoint subspaces of **V**, meaning **X** $\cap$ **Y** $\cap$ **Z** = 0.

Thus, **x** $\in$ **X**, **y** $\in$ **Y** and **z** $\in$ **Z** are also disjoints, meaning they are independent of each other as they have nothing in common.

Therefore, the **linear combination** (addition of scalar multiples) of **x**, **y**, **z** will be zero if and only if the scalar multiples are zero.

$$ax + by + cz = 0 \; iif \; a, b, c = 0$$

Linear Span and Basis

The linear combination of linearly independent vectors from disjoint subspace can also form a Linear Space.

The **Linear Space** formed by the linear combination of linearly independent vectors is known as **Linear Span**, and those linearly independent vectors are known as **Basis**.

Suppose **X** is the **Linear Span** formed by the **Basis** { $\mathbf{x}_1$, $\mathbf{x}_2$, $\mathbf{x}_3$, … , $\mathbf{x}_n$ } such that:

$$\sum_{i=1}^{n} a_i x_i = 0 \; iff \; a_i = 0$$

Multiplication in Linear Space

Although **Linearity** explicitly defines **Linear Combination** by **Addition** and **Scalar Multiplication**, it does not explicitly define **Multiplication**.

$$+(u, v) \rightarrow u + v$$

$$.(a, u) \rightarrow au$$

Therefore, there are many **Multiplication Rules** in Linear Space; however, Multiplication in Linear Space is mostly determined by two important aspects: **Duality** and **Symmetry**.

$$\times (u, v) \rightarrow \cdots$$

Orthogonality

Orthogonality in Linear Space is actually defined from the Linear Independence.

What would happen if we **multiply** two **linearly independent** entities in Linear Space?

To answer this, we have to define the Multiplication of **Basis** { $\mathbf{g}_1$, $\mathbf{g}_2$, $\mathbf{g}_3$, … , $\mathbf{g}_n$ } such that:

$$g_i . g_j = \delta_{ij}$$

δ_{ij} is known **Krönecker Delta** symbol:

$$\delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

From the perspective of Orthogonality, the multiplication of two bases will be defined by **Krönecker Delta** .

Symmetry

Suppose **X** is the **Linear Space** formed by the **Basis** { $\mathbf{x}_1$, $\mathbf{x}_2$, $\mathbf{x}_3$, … , $\mathbf{x}_n$ }.

The **rotation** of the basis can be defined by **different permutation** of order pairs of basis such that:

$$\{\{ \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots , \mathbf{x}_n \}, \dots , \{ \mathbf{x}_n, \mathbf{x}_{n-1}, \mathbf{x}_{n-2}, \dots , \mathbf{x}_1 \}\}$$

If any two different order pairs of permutation are **equal**, then the basis is said to be **symmetric**. Thus, the **order** does matter in **Linear Space**, unlike a **Set**.

Levi-Civita symbol defines the symmetry of the basis by defining rotating permutations:

$$\epsilon_{1\cdots n} = \begin{cases} 1, & \text{Even Permutation} \\ -1, & \text{Odd Permutation} \\ 0, & \text{Same Permutation} \end{cases}$$

Vector Space

Now, we can formally defined a Vector Space.

A Vector Space **V** is the **Linear Span** formed by the **Basis** { $\mathbf{g}_1$, $\mathbf{g}_2$, $\mathbf{g}_3$, … , $\mathbf{g}_n$ } such that:

$$\sum_{i=1}^{n} v_i \boldsymbol{g}_i = 0 \; iff \; v_i = 0$$

Any vector in the Vector Space can be uniquely identified by the Linear Combination of the Basis Vectors of the Vector Space.

Therefore, a **Vector** has two aspect: **components** and **basis**:

$$\boldsymbol{v} = \sum v_i \boldsymbol{g}_i$$

Dual Space

Suppose **V** is a **Vector Space** defined by the basis the **Basis** { $\mathbf{g}_1$, $\mathbf{g}_2$, $\mathbf{g}_3$, … , $\mathbf{g}_n$ }.

$$v = v_1 \boldsymbol{g}_1 + \cdots + v_n \boldsymbol{g}_n$$

Then, we can define a function $l: V \rightarrow R$ such that:

$$g_i^*(\boldsymbol{g}_j) = \delta_{ij}$$

Therefore, this function will return the components of the Vector, given the **Index of a basis** such that:

$$g_i^*(\boldsymbol{v}) = v_i$$

Then, the set of all such linear functions is defined as V*, a **Dual Space** of **V**, as such v_i is the **Covector** of **v**, and g^*_i are **Dual Basis** of $\mathbf{g_i}$.

In this sense, a vector **v** is made up of **Duo**: **Covector** g*(**v**) and **Basis g**.

Inner Product

Now, we can define one of the most important **multiplication rules** in Vector Space; however, please let me warn you **not to take it granted** for **Multiplication** in Vector Space.

Suppose we have two vectors **u** and **v**, one form of multiplication called **Inner Product** is defined by **Orthogonality** and **Duality** of Vector Space, such that:

$$.(u,v) \rightarrow u.v = \sum_{i=1}^{n} u_i v_i (\boldsymbol{g}_i . \boldsymbol{g}_i)$$

According to **Duality**, the Inner Product is defined as the Sum of the Product of Individual **Covector** g*(**v**) and **Basis g**.

However, according to **Orthogonality**, the Product of ($\mathbf{g}_i$. $\mathbf{g}_i$) will yield a 1 while ($\mathbf{g}_i$. $\mathbf{g}_j$) will yield a 0.

Example of Inner Product

Let us consider two vectors **u** and **v** in 3 Dimensional Vector Space V.

$$\boldsymbol{v} = v_1\boldsymbol{g}_1 + v_2\boldsymbol{g}_2 + v_3\boldsymbol{g}_3$$

$$\boldsymbol{u} = u_1\boldsymbol{g}_1 + u\boldsymbol{g}_2 + u\boldsymbol{g}_3$$

Then, the **Inner Product** of **u** and **v** is:

$$\begin{aligned}\mathbf{u}.\mathbf{v} = {} & \mathrm{u}_1\mathrm{v}_1\,(\mathbf{g_1}.\mathbf{g_1}) + \mathrm{u}_1\mathrm{v}_2\,(\mathbf{g_1}.\mathbf{g_2}) + \mathrm{u}_1\mathrm{v}_3\,(\mathbf{g_1}.\mathbf{g_3}) + \\ & \mathrm{u}_2\mathrm{v}_1\,(\mathbf{g_2}.\mathbf{g_1}) + \mathrm{u}_2\mathrm{v}_2\,(\mathbf{g_2}.\mathbf{g_2}) + \mathrm{u}_2\mathrm{v}_3\,(\mathbf{g_2}.\mathbf{g_3}) + \\ & \mathrm{u}_3\mathrm{v}_1\,(\mathbf{g_3}.\mathbf{g_1}) + \mathrm{u}_3\mathrm{v}_2\,(\mathbf{g_3}.\mathbf{g_2}) + \mathrm{u}_3\mathrm{v}_3\,(\mathbf{g_3}.\mathbf{g_3}) + \end{aligned}$$

$$= \mathrm{u}_1\mathrm{v}_1 + \mathrm{u}_2\mathrm{v}_2 + \mathrm{u}_3\mathrm{v}_3$$

Now, we get the answer we are usually familiar with. However, it is only possible because of **Duality** and **Orthogonality** of Vector Space.

What if the Orthogonality is not defined?

Inner Product Space

Inner Product Space is actually a Vector Space for which Inner Product is defined.

However, to be able to define **Inner Product**, the following condition must exist.

$$g_i . g_j = \delta_{ij}$$

Inner Product Space can define a general Map such that:

$$g : V \times V \rightarrow R$$

Actually, **Inner Product Space** can still be defined even without Orthogonal Condition.

However, in such case, the product of the basis is no longer **Krönecker Delta**. Instead, it will become a **Tensor**, commonly known as **Metric Tensor**. Therefore, Inner Product Space is dependent on how Metric Tensor is defined.

$$g_i . g_j = T$$

Inner Product Space

Now let us define the generic inner product of basis.

$$g_i . g_j = T$$

If we list all the pairs of $\mathbf{g}_i . \mathbf{g}_j$, we get:

$$\{(g_1 . g_1), (g_1 . g_2), \cdots, (g_n . g_n)\}$$

And if we arrange in Matrix Form, we get:

$$T = \begin{pmatrix} g_1 . g_2 & \cdots & g_1 . g_n \\ \vdots & \ddots & \vdots \\ g_n . g_2 & \cdots & g_n . g_n \end{pmatrix}$$

If we can define Orthogonality, **T** = Identity Matrix, **I**.

$$T = g_i . g_j = \delta_{ij} = I$$

Change of Basis

With the definition of Inner Product Space, we now will be able to transform the basis { $\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{e}_3$,…, $\mathbf{e}_n$} of a vector **v** to { $\mathbf{f}_1$, $\mathbf{f}_2$, $\mathbf{f}_3$,…, $\mathbf{f}_n$}.

$$\boldsymbol{v} = \sum v_i \boldsymbol{e}_i = \sum v'_i \boldsymbol{f}_i$$

Vectors are invariant, meaning they do not change even though their basis (coordinates) and components are changing.

By defining **Metric Tensors**, we can actually transform from $\mathbf{e}_i$ to $\mathbf{f}_j$, such that:

$$f_i = \sum A_{ij} e_j \qquad A = f_i . e_j$$

$$e_i = \sum B_{ij} f_j \qquad B = e_i . f_j$$

$$AB = BA = \delta_{ij} = I$$

Change of Basis

However, when transform a vector v, we not only have to transform the basis, but also have to transform vector components as well.

But let us start with the **transformation of the basis**:

$$f_i = \sum A_{ij} e_j$$

For the **transformation of the components**:

$$\sum v'_i \boldsymbol{f_i} = \sum v_i \boldsymbol{e_i} \qquad \text{But, } e_i = \sum B_{ij} f_j$$

$$\sum v'_i \boldsymbol{f_i} = \sum v_i B_{ij} \boldsymbol{f_j}$$

$$v'_j = \sum v_i B_{ij}$$

Change of Basis

As we can see, the transformation of the basis is related by Forward Tensor, while the transformation of the components is related by Backward (Inverse) Tensor.

Therefore, **Forward Transformation** is known as **Covariant**:

$$f_i = \sum A_{ij} e_j$$

And **Backward (Inverse) Transformation** is known as **Contravariant** :

$$v'_j = \sum v_i B_{ij}$$

Thus, for all transformation, there will be both **Covariant** and **Contravariant Transformation Rules**.

Tensor Notation

Generally speaking, **Tensors** are usually represented with **Upper Indices** to represent **Contravariant** Components and **Lower Indices** to represent **Covariant** Components, such that:

$$Tensor = T^i_j$$

where **i** represents the indices for **Contravariant** Components and **j** represents the indices for **Covariant** Components.

Now, we will represent a vector in **Tensor Notation** such that:

$$\boldsymbol{v} = v^i \boldsymbol{g}_i$$

$$\boldsymbol{v}.\boldsymbol{u} = v^i u^j \delta_{ij}$$

Raising or Lowering Indices

If we have a vector **v** with the basis $\mathbf{g_j}$, **Duality** is defined such that:

$$g^i(\boldsymbol{g}_j) = \delta^i_j$$, where g^i is the **dual basis** of $\mathbf{g_j}$.

Then, **Orthogonality** is defined such that:

$$\boldsymbol{g}_i . \boldsymbol{g}_j = g_{ij} = \delta_{ij}$$, where g_{ij} is the **Metric Tensor**

Therefore, the **Metric Tensor** for **dual basis**, which is the **Inverse Metric Tensor**, can be defined as:

$$g^i . g^j = g^{ij}$$

Thus,

$$g^{ij} g_{jk} = \delta^i_k$$

Therefore, we can raise or lower the indices in Tensor Notation by Metric Tensor or Dual Metric Tensor.

Change of Indices

Suppose, we have a Tensor A^j , and g_{ij} is the **Metric Tensor**:

$$A_i = g_{ij}A^j$$

Likewise, if we have a Tensor A_j , and g^{ij} is the **Inverse Metric Tensor**:

$$A^i = g^{ij}A_j$$

Example of Metric Tensor

One of the famous Metric Tensors in Physics is **Minkowski Metric Tensor**, which is normally used in **Special Theory of Relativity**.

$$\eta = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

In **Minkowski** Space, the space is defined by 4 dimensional basis {t, x, y, z}, and instead of Krönecker Delta **δ**, the Metric Tensor is defined by **η**.

Therefore, the Inner Product in **Minkowski** Space would be:

$$\boldsymbol{v}.\boldsymbol{u} = v^i u^j \eta_{ij}$$

Normed Space

From the definition of Inner Product Space, we now can define another space called **Normed Space**.

$$\boldsymbol{v}.\boldsymbol{v} = v^i v^j T_{ij}$$, where T_{ij} is the **Metric Tensor**

Now, Norm of Inner Product is defined as: $\|.\| : V \times V \rightarrow R$

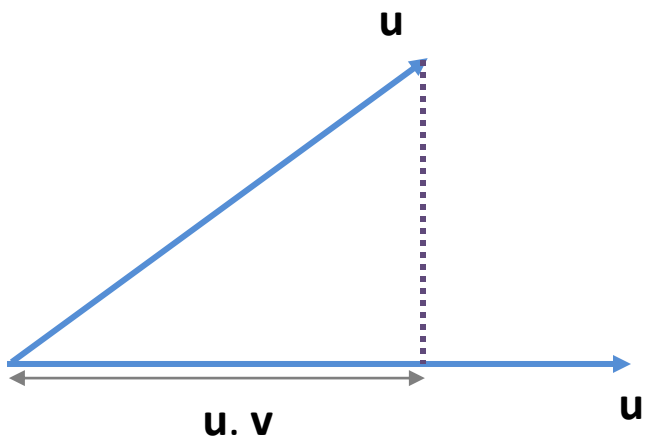
$$\|\boldsymbol{v}.\boldsymbol{v}\| = \left(v^i v^j T_{ij}\right)^{\frac{1}{2}}$$

Therefore, the distance between two vectors is: $\boldsymbol{r} = \boldsymbol{u} - \boldsymbol{v}$

$$\|\boldsymbol{r}.\boldsymbol{r}\| = \left(r^i r^j T_{ij}\right)^{\frac{1}{2}}$$

Cauchy–Schwarz Inequality

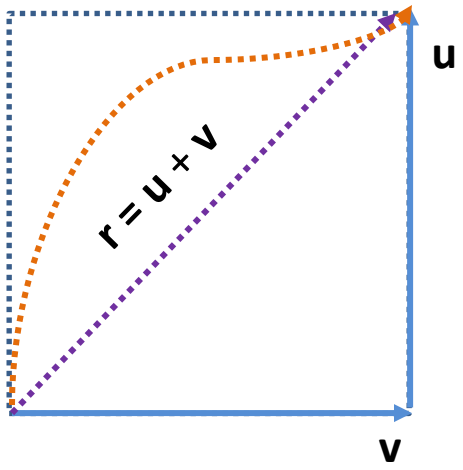
Cauchy–Schwarz Inequality states that:



$$|\boldsymbol{u}.\boldsymbol{v}| \leq \|\boldsymbol{u}\|\|\boldsymbol{v}\|$$

Triangle Inequality

Similar to **Cauchy–Schwarz Inequality**, **Triangle Inequality** can be defined as:



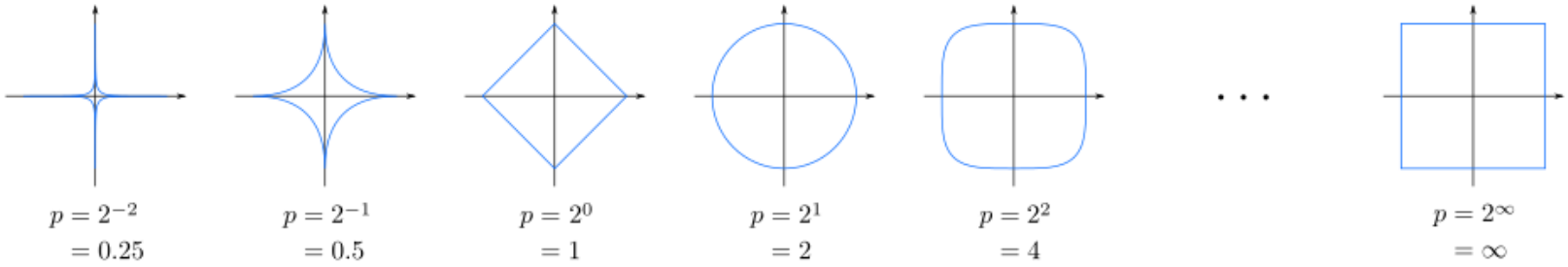
$$\|\boldsymbol{u}+\boldsymbol{v}\| \leq \|\boldsymbol{u}\| + \|\boldsymbol{v}\|$$

Depending on **Metric Tensor** T_{ij}, we can have different distance between **u** and **v**. However, the **maximum distance** will always be less than and equal to the sum of the length of **u** and **v**.

Minkowski Distance

Minkowski Distance is the generalization of different **Distance Measure**, such as Euclidean Distance, Manhattan Distance, and so on.

Measuring distance is not so straight-forward, as the space itself can be a curve.

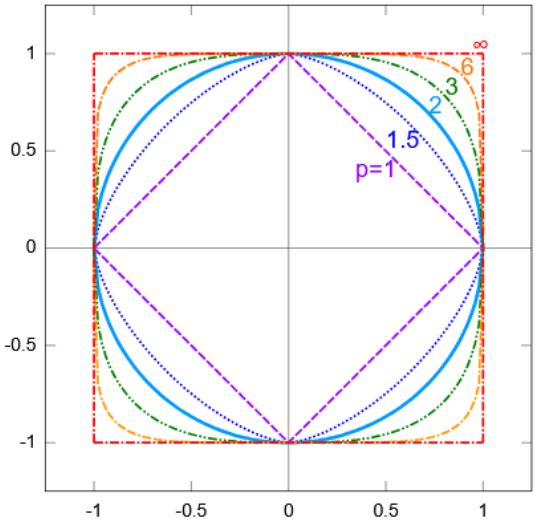


$$\|\boldsymbol{u} - \boldsymbol{v}\|_p = \lim_{p \to \infty} \left(\sum_{i=1}^{n} (u_i - v_i)^p \right)^{\frac{1}{p}}$$

P-Norm

P-Norm is the generalization of **Distance Measure** between two vectors **v** and **u**, given different **Metric Tensor** T_{ij}.

We usually use different **P-Norm** in **Machine Learning**, and actually, different P-Norms correspond to different Metric Tensor. When P = 2, it has become **Euclidean Distance** for which the Metric Tensor is **Krönecker Delta**.



$$\|\boldsymbol{u}-\boldsymbol{v}\|_p = \left(\sum_{i=1}^{n}(u_i - v_i)^p\right)^{\frac{1}{p}}$$

Set and Vector

At this point, it should naturally occurs to us as to what is the difference between **Set** and **Vector**, as both of which can represent **State Space**.

A **Set** is a mathematical concept to represent a list of **distinct entities** with common **Properties**. Although a **Set** can represent **Properties** of its elements, it cannot represent their **Structure**.

However, unlike a **Set**, a **Vector** can represent **Structure** of its elements by retaining their **Shape**. Thus, a **Vector** is a **Structured Set**.

Vectors are **invariant** in a sense that their **Structure** does not change under **Linear Transformation**. Of course, their structure still changes under **Non-Linear Transformation**.

In a nut shell, a **Set** only defines **Properties**, while a **Vector** additionally represents **Structures** or **Patterns**.

Linear Map

Now, let us look into how the **Structure** of a **Vector** changes under **Linear Transformation**.

Mathematically, such **Structural Transformation** is defined as **Morphism**.

Now, let us formally define a Linear Map $T: \boldsymbol{V} \rightarrow \boldsymbol{U}$, we call such *Linear Map* as Vector Space **Homomorphism**.

The inverse of Linear Map T is $T^{-1}: \boldsymbol{U} \rightarrow \boldsymbol{V}$, and if T^{-1} exist for T, then T is **bijective**, and such *Linear Map* called Vector Space **Isomorphism**.

Representation of Linear Map

Suppose we have a vector space $\mathbf{v} \in \mathbf{V}$ with the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \ldots, \mathbf{e}_n\}$ and a vector space $\mathbf{u} \in \mathbf{U}$ with the basis $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \ldots, \mathbf{f}_m\}$.

Now, T is Linear Map such that $T: \boldsymbol{V} \rightarrow \boldsymbol{U}$

Then we can represent **u** in terms of **v** through a Linear Map T such that:

$$\boldsymbol{u} = T\boldsymbol{v}$$

$$\boldsymbol{u} = T\left(\sum_{i=1}^{n} v^i \boldsymbol{e}_i\right)$$

$$\boldsymbol{u} = \sum_{i=1}^{n} v^i T\boldsymbol{e}_i$$

Representation of Linear Map

Now, let us see how the **basis** are transformed by the Linear Map T.

Since $T\mathbf{e}_i \in \mathbf{U}$ with the basis $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \ldots, \mathbf{f}_m\}$, we can now represent $T\mathbf{e}_i$ in terms of $\mathbf{f}_j$.

$$T\boldsymbol{e}_i = \sum_{j=1}^{m} T_{ij}\boldsymbol{f}_j$$

$$T_{ij} = \boldsymbol{f}_i . T\boldsymbol{e}_j$$

Representation of Linear Map

Again, let us see how the **components** are transformed by the Linear Map T.

$$\boldsymbol{u} = \sum_{i=1}^{n} v^i T \boldsymbol{e}_i$$

$$\sum_{j=1}^{m} u^j \boldsymbol{f}_j = \sum_{i=1}^{n} v^i T_{ij} \boldsymbol{f}_j \qquad \text{But, } T\boldsymbol{e}_i = \sum_{j=1}^{m} T_{ij} \boldsymbol{f}_j$$

$$u^j = \sum_{i=1}^{n} v^i T_{ij}$$

Representation of Linear Map

Linear Transformation of the components and the basis:

$$T\boldsymbol{e}_i = \sum_{j=1}^{m} T_{ij}\boldsymbol{f}_j$$

$$v^j = \sum_{i=1}^{n} u^i T_{ij}$$

Now T_{ij} is the **m x n** Matrix, which is the Matrix Representation of a Linear Map:

$$T_{ij}: \boldsymbol{V} \rightarrow \boldsymbol{U}$$

Change of Basis in Linear Map

Now, we will try to represent a generic linear map by defining change of basis as well.

As we already know, the change of basis is defined by two metric tensors: **forward** and **backward** metric tensors.

$$f_i = \sum A_{ij} e_j \qquad A = f_i . e_j$$

$$e_i = \sum B_{ij} f_j \qquad B = e_i . f_j$$

Change of Basis in Linear Map

Suppose we have a vector space $\mathbf{v} \in \mathbf{V}$ with the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \ldots, \mathbf{e}_n\}$ and a vector space $\mathbf{u} \in \mathbf{U}$ with the basis $\{\mathbf{g}_1, \mathbf{g}_2, \mathbf{g}_3, \ldots, \mathbf{g}_m\}$.

But we will apply change of basis from $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \ldots, \mathbf{e}_n\}$ to $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \ldots, \mathbf{f}_n\}$ and $\{\mathbf{g}_1, \mathbf{g}_2, \mathbf{g}_3, \ldots, \mathbf{g}_m\}$ to $\{\mathbf{h}_1, \mathbf{h}_2, \mathbf{h}_3, \ldots, \mathbf{h}_m\}$.

The linear map T from **V** to **U** is defined as: $T_{ij}: \boldsymbol{V} \rightarrow \boldsymbol{U}$

$$T\boldsymbol{e}_i = \sum_{j=1}^{m} T_{ij}\boldsymbol{g}_j \quad T_{ij} = \boldsymbol{g}_i . T\boldsymbol{e}_j$$

However, we have 4 metric tensors: A, A^{-1} = B, C, C^{-1} = D

$$\boldsymbol{f}_i = \sum_{j=1}^{n} A_{ij}\boldsymbol{e}_j \quad \boldsymbol{e}_i = \sum_{j=1}^{n} B_{ij}\boldsymbol{f}_j \quad \boldsymbol{h}_i = \sum_{j=1}^{m} C_{ij}\boldsymbol{g}_j \quad \boldsymbol{g}_i = \sum_{j=1}^{m} D_{ij}\boldsymbol{h}_j$$

Change of Basis in Linear Map

Since the new linear map is:

$$T\boldsymbol{f}_i = \sum_{j=1}^{m} T'_{ij}\boldsymbol{h}_j$$

$$TA_{ij}\boldsymbol{e}_i = \sum_{j=1}^{m} T'_{ji}C_{ij}\boldsymbol{g}_j$$

$$C_{ji}^{-1}TA_{ij}\boldsymbol{e}_i = \sum_{j=1}^{m} T'_{ij}\boldsymbol{g}_j$$

$$\therefore\ T'_{ij} = C_{ia}^{-1}T_{ab}A_{bj}$$

$$T'_{ij} = D_{ia}T_{ab}A_{bj}$$

Linear Decomposition

Suppose we have a vector space $\mathbf{v} \in \mathbf{V}$ with the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \ldots, \mathbf{e}_n\}$ and a vector space $\mathbf{u} \in \mathbf{U}$ with the basis $\{\mathbf{g}_1, \mathbf{g}_2, \mathbf{g}_3, \ldots, \mathbf{g}_m\}$, and there is a linear map T:

$$T_{ij}: \boldsymbol{V} \rightarrow \boldsymbol{U}$$

As we can see, a generic linear map can be **factorized** with the **Inverse Metric Tensor** for the basis $\mathbf{g}_j$ and the **Metric Tensor** for the basis $\mathbf{e}_i$.

$$T'_{ij} = D_{ia} T_{ab} A_{bj}$$

Actually, it is not just a coincidence that there are many similar **Linear Decompositions** by **Factorization**, such as Singular Value Decomposition, Eigen Value Decomposition.

Vector Space of Linear Map

Suppose we have a vector space $\mathbf{v} \in \mathbf{V}$ with the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \ldots, \mathbf{e}_n\}$ and a vector space $\mathbf{u} \in \mathbf{U}$ with the basis $\{\mathbf{g}_1, \mathbf{g}_2, \mathbf{g}_3, \ldots, \mathbf{g}_m\}$, and there is a linear map T:

$$T: V \to U$$

However, let us consider a set of all linear maps **L** from **V** to **U**:

$$L: V \to U \quad \text{, where } T \in L$$

It turns out that **L** is also a Linear Space or Vector Space.

$$\alpha(S + T)\boldsymbol{v} = \alpha S\boldsymbol{v} + \alpha T\boldsymbol{v} \quad \text{, where } S, T \in L$$

Outer Product

Suppose we have a vector space $\mathbf{v} \in \mathbf{V}$ with the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \ldots, \mathbf{e}_n\}$ and a vector space $\mathbf{u} \in \mathbf{U}$ with the basis $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \ldots, \mathbf{f}_m\}$.

Then, we can define a new multiplication rule called **Outer Product** such that:

$$(\boldsymbol{f}_j \otimes \boldsymbol{e}_i)\boldsymbol{v} = (\boldsymbol{v}.\boldsymbol{e}_i)\boldsymbol{f}_j = v^i \boldsymbol{f}_j$$

As we can see, it is the basis of Linear Map, such that:

$$\boldsymbol{f}_j \otimes \boldsymbol{e}_i : \boldsymbol{V} \rightarrow \boldsymbol{U}$$

Then, we can represents a Linear Map $T: V \rightarrow U$, with the Linear Combination of $(\mathbf{f}_j \otimes \mathbf{e}_i)$:

$$T = \sum T_{ij} (\boldsymbol{f}_j \otimes \boldsymbol{e}_i)$$

Outer Product Space

Suppose we have a vector space $\mathbf{v} \in \mathbf{V}$ with the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \ldots, \mathbf{e}_n\}$ and a vector space $\mathbf{u} \in \mathbf{U}$ with the basis $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \ldots, \mathbf{f}_m\}$.

Then, the **Linear Span** of $(\mathbf{f}_j \otimes \mathbf{e}_i)$ defines the **Outer Product Space**, which is the Vector Space of Linear Map L:

$$L: V \rightarrow U \text{ , where } L = Linear\ Span\{(\boldsymbol{f}_j \otimes \boldsymbol{e}_i)\}$$

Now, how could we understand Outer Product Space more intuitively?

$$T'_{ij} = D_{ia} T_{ab} A_{bj}$$

As we can see, the Linear Map is the factorization of two Metric Tensors: the Inverse Metric Tensor of the basis $\mathbf{f}_j$ and the Metric Tensor of the basis $\mathbf{e}_i$.

Therefore, the outer product space specifies the **relationship** between such Metric Tensors.

Linear Decomposition

We will look into one of the famous **Linear Decompositions**, called **Spectral Decomposition**. However, before we look into Spectral Decomposition, we will define some Identities first.

$$e^{i\omega} = \cos(\omega) + isin(\omega)$$

As we can see, the derivative of $e^{\lambda x}$ is:

$$\frac{d}{dx} e^{\lambda x} = \lambda e^{\lambda x}$$

And, **Fourier Series** is defined as:

$$\sum_{k=1}^{n} c_k e^{i\omega_k} = \sum_{k=1}^{n} a_k \sin(\omega) + b_k icos(\omega)$$

Spectral Decomposition

If a Linear Map T transform a vector space $\mathbf{v} \in \mathbf{V}$ with the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \ldots, \mathbf{e}_n\}$ onto its Scalar Multiple, then:

$$T\boldsymbol{v} = \lambda \boldsymbol{v}$$

Such a vector is called **Eigen Vector**, the components of T_{ij} are called **Eigen Values**.

Spectral Decomposition, also known as **Eigen Decomposition,** is one of the mostly used Linear Decompositions.

At any rate, we can conclude that the nature of **Eigen Vector** must be **Exponential Function**, and the nature of **Linear Map** must be **Derivatives**.

$$\frac{d}{dx} e^{\lambda x} = \lambda e^{\lambda x}$$

Spectral Decomposition

Suppose Linear Map T represents a Linear System such that:

$$\boldsymbol{y} = T\boldsymbol{x}$$

Now, the input **x** can be defined as a vector **x** with the basis of $\{e^{i\omega 1}, e^{i\omega 2}, e^{i\omega 3}, \ldots, e^{i\omega n}\}$ such that:

$$x = \sum_{k=1}^{n} c_k e^{i\omega_k}$$

Since a vector **x** is an Eigen Vector, the output vector will be scalar multiple of **x** itself.

$$\lambda \boldsymbol{x} = T\boldsymbol{x}$$

Actually, through Eigen Values, we can understand the Nature of Linear Systems.

As the input vector is the **Linear Combinations of Sinusoidal**, we can analyze how different components of Input Vector change under the Linear Transformation of Linear Map T.

Spectral Decomposition

Since

$$\lambda \boldsymbol{x} = T\boldsymbol{x}$$

From this polynomial equation, also known as **Characteristic Equation**:

$$p(\lambda) = \det(T - I\lambda)$$

we can solve for the Eigen Values.

$$\lambda_i \boldsymbol{x}_i = \sum_{j=1}^{n} T_{ij} \boldsymbol{x}_j$$

Spectral Decomposition

Once we get the Eigen Values and Eigen Vectors, we can factorize T into Spectral Decomposition such that:

$$T = Q\Lambda Q^{-1}$$

Q is the n x n Matrix whose ***i***th column is the eigenvector $\boldsymbol{q}_i$ of T, and Λ is the diagonal matrix whose diagonal elements are the corresponding Eigen Values, $\Lambda_{ii} = \lambda_i$.

Now, if we look from the perspective of the previous Linear Map Factorization, what can we conclude?

$$T'_{ij} = D_{ia} T_{ab} A_{bj}$$

Inner and Outer Product Space

Suppose we have a vector space $\mathbf{v} \in \mathbf{V}$ with the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \ldots, \mathbf{e}_n\}$ and a vector space $\mathbf{u} \in \mathbf{U}$ with the basis $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \ldots, \mathbf{f}_m\}$.

Inner Product Space is defined by Inner Product such that:

$$\boldsymbol{e}_i . \boldsymbol{e}_j \rightarrow T_{ij} \quad \text{or} \quad \boldsymbol{e}_i . \boldsymbol{f}_j \rightarrow G_{ij}$$, where T_{ij} and G_{ij} are Metric Tensors

Again, **Outer Product Space** is defined by Outer Product such that:

$$\boldsymbol{f}_j \otimes \boldsymbol{e}_i : \boldsymbol{V} \rightarrow \boldsymbol{U}$$, where $\mathbf{f_j} \otimes \mathbf{e_i}$ is the basis of Linear Span $\{\mathbf{f_j} \otimes \mathbf{e_i}\}$

Any Linear Span of Outer Product Space can also be factorized by Metric Tensors of corresponding Vector Spaces **V** and **U**, which actually represents the **latent relationship** of Metric Tensors.

$$T'_{ij} = B^{-1}_{ix} T_{xy} A_{yj}$$

Linear Forms

Before proceeding to **Tensors**, let me introduce one of the most important concepts in Linear Space, known as **Linear Forms**.

Why Linear Forms are particularly important is mainly because the concepts of Linear Forms will be extended to **K-Linear Forms** and **K-Differential Forms**, which are of primary importance in Tensors.

Without further ado, let us look into Linear Forms.

Linear Forms

Suppose we have a vector space $\mathbf{v} \in \mathbf{V}$ with the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \ldots, \mathbf{e}_n\}$. Then, we can represent a vector **v** as a Linear Combinations of the Basis $\mathbf{e}_i$ such that:

$$\boldsymbol{v} = a_1\boldsymbol{e}_1 + a_2\boldsymbol{e}_2 + a_1\boldsymbol{e}_3 + \cdots + a_n\boldsymbol{e}_n$$

We can actually define a Linear Function $l: \boldsymbol{V} \rightarrow R$ such that:

$$l(\boldsymbol{V}) = V^*$$

$$l(e_i) = a_i$$

And such Linear Function $l: \boldsymbol{V} \rightarrow R$, is known as **Linear Form** or **Linear 1-Form**.

Actually, Linear Form is also a Linear Function between Vector Space **V** and Dual Space **V***.

$$g_i^*(\boldsymbol{v}) = v_i$$

Example of Linear Forms

Suppose we have a vector space **v** ∈ **V** with the basis {**x, y, z**}. Then, we can represent a vector **v** as a Linear Combinations of the Basis {**x, y, z**} such that:

$$\boldsymbol{v} = a_1\boldsymbol{x} + a_2\boldsymbol{y} + a_1\boldsymbol{z}$$

Then, Linear Function $l: \boldsymbol{V} \rightarrow R$ is defined such that:

$$f(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}) = R = a_1 x + a_2 y + a_1 z$$

If (x, y, z) are the co-ordinates (**unit direction**) such that {(1, 0, 0), (0, 1, 0), (0, 0, 1)}, then

$$f(1,0,0) = a_1$$
$$f(0,1,0) = a_2$$
$$f(0,0,1) = a_3$$

Such Linear Function is also called **Evaluation Function**, because it can tell us something about vector **v**. If we provide a **directional coordinates**, it will give us the **components of vector in that direction**, which is extremely important to analyze vectors.

Example of Differential Forms

Now, we will extend the concept of **Linear 1-Forms** to represent **Differential 1-Forms**.

Suppose we have a vector field defined by the **Gradient of Scalar Field ∇F**(x, y) with the basis {**i, j**}. Then, we can represent a gradient vector field **∇F** as a Linear Combinations of the Basis {**i, j**} such that:

$$\nabla F(x,y) = \frac{\partial F}{\partial x}\boldsymbol{i} + \frac{\partial F}{\partial y}\boldsymbol{j}$$

Then, the **Differential Form** *dF* is defined such that:

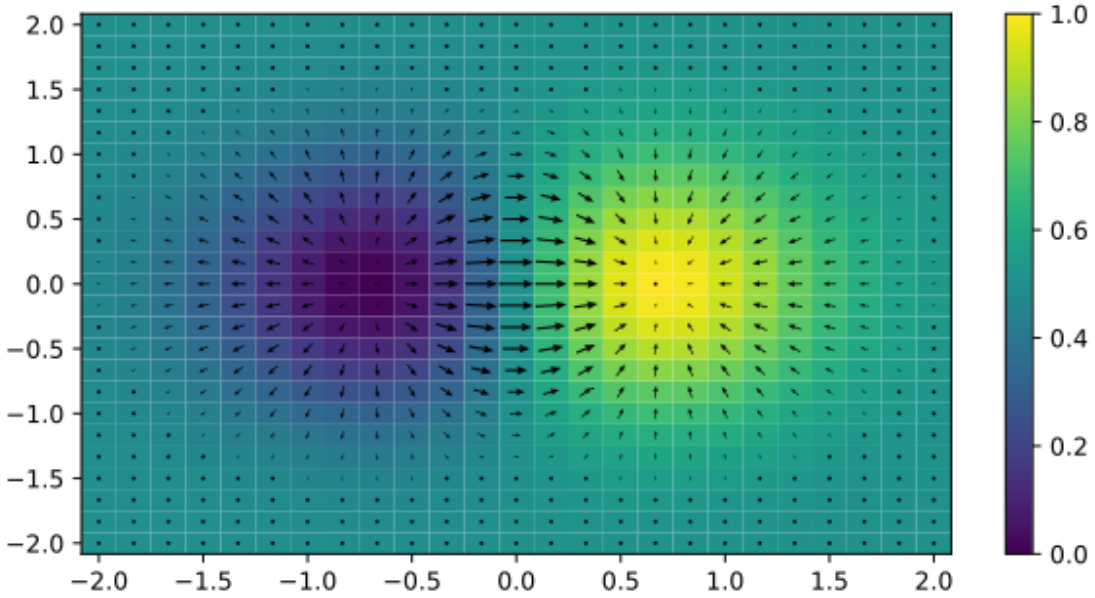
$$\mathrm{dF} = \frac{\partial F}{\partial x}dx + \frac{\partial F}{\partial y}dy$$

As we can see while **Gradient ∇F** represents Vector Field, known as **Gradients**, **Differential Form** *dF* represents Covector Field, known as **Contours**. Of course, we will look more into Differential Forms later.

Vector and Covector Fields

Now, let us visualize **Vector Fields** and **Covector Fields**, or **Gradients** and **Contours**. As we can see, **Gradients** and **Contours** are *orthogonal*, which makes sense as they are Dual Spaces.

$$\nabla F(x,y) = \frac{\partial F}{\partial x}\boldsymbol{i} + \frac{\partial F}{\partial y}\boldsymbol{j}$$



$$\mathrm{dF} = \frac{\partial F}{\partial x}dx + \frac{\partial F}{\partial y}dy$$

Tensor

At this point, it should naturally occurs to us as to what really are Tensors.

Tensors are in fact **Multi-Linear Map**, which can relate different **Multiple Linear Spaces**.

Multi-Linearity can be defined as **K-Linearity**, which involves different **K** Linear Spaces: { $\mathbf{V_1}$, $\mathbf{V_2}$, $\mathbf{V_3}$, …, $\mathbf{V_k}$ }, which can be linearly related **if others are kept constant**.

From the concept of **Linear Forms**, we can actually define **Tensors** with **K-Linear Forms**.

Now, let us formally define **Tensors**.

Tensor

Let **V** be a N Dimensional Space with the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \ldots, \mathbf{e}_n\}$ and let $\mathbf{V}^k$ be a set of k-tuples $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \ldots, \mathbf{v}_k)$ where $\mathbf{v}_k \in \mathbf{V}$. Then, a function is defined as:

$$T: \boldsymbol{V}^{\boldsymbol{k}} \to R \quad \text{, which is } T: \boldsymbol{V} \times \boldsymbol{V} \cdots \times \boldsymbol{V} \to R$$

T is a Multi-Linear Function which represents the multi-linear relationship between $\mathbf{V}^k$ Vector Spaces, then *T* is said to be **K-Linear** or **K-Tensor**. Please remember **K-Linear** is **NOT Linear**; however **1-Linear** is **Linear**.

In **K-Linear Forms**, we can write T as:

$$T = L^k(V)$$

As we can see, **1-Linear Form** is:

$$L^1(V) \to V^*$$

And **0-Linear Form** is just a scalar:

$$L^0(V) \to R$$

Ranks of Tensor

With **K-Linear Form**, we can define the Ranks of a Tensor T such that T is **Rank K** Tensor.

$$T = L^k(V)$$

As such, 2-Linear (Bi-Linear) Form is Rank 2 Tensor, which is a Matrix, 1-Linear Form is Rank 1 Tensor, which is a Covector, and 0-Linear Form is Rank 0 Tensor, which is a Scalar.

In addition, with **Tensor Notation**, which can also represents **Covariant** and **Contravariant Components**:

$$T^i_{jk} \quad \text{or} \quad T^{ab}_d$$

We can also call T^i_{jk} is a Rank 1-2 Tensors which is Rank 3 Tensor in K-Linear Form, or T^{ab}_d is a Rank 2-1 Tensor, which is also a Rank 3 Tensor in K-Linear Form.

Multi-Linearity

Multi-Linearity means that some components are linear if only if other components are kept constant. Thus we can define **Multi-Linearity** as :

$$f(v_1, \cdots, u_i + av_i, \cdots, v_n) = f(v_1, \cdots, u_i, \cdots, v_n) + af(v_1, \cdots, v_i, \cdots, v_n)$$

Suppose we have a **non-linear** function, which is defined such that:

$$f(x, y) = xy$$

Then, let us define

$$\begin{aligned} f((x + au), y) &= (x + au)y \\ &= xy + auy \\ &= f(x, y) + af(u, y) \end{aligned}$$

As we can see, Multi-Linearity is **Not Linear**, as *f*(x, y) is Multi-Linear, but it is not Linear.

Multi-Linear Representation

Suppose we have a vector space $\mathbf{v} \in \mathbf{V}$ with the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \ldots, \mathbf{e}_n\}$. Then, we can represent:

$$\boldsymbol{v} = \sum_{i=1}^{n} v^i \boldsymbol{e}_i$$

However, if we have tuples of $\mathbf{v}_i \in \mathbf{V}$ with the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \ldots, \mathbf{e}_n\}$, then,

$$\boldsymbol{v_i} = \sum_{j=1}^{n} v^{ij} \boldsymbol{e}_{ij}$$

Multi-Linear Representation

In **Linear Form**, we can defined $f(\mathbf{v})$ with the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \ldots, \mathbf{e}_n\}$ for **directional coordinates**, such as {(1, 0, …, 0), … , (0, 0, …, 1)}, as :

$$f(\boldsymbol{v}) = f\left(\sum_{i=1}^{n} v^i \boldsymbol{e}_i\right) = \sum_{j=1}^{n} A_j v_j$$

Thus, if we have k-tuples of $\boldsymbol{v}_k$ in **K-Linear Form** of $T^k(\mathbf{V}) = f(\mathbf{v}_1, \ldots, \mathbf{v}_k)$, then,

$$f(\boldsymbol{v}_1, \cdots, \boldsymbol{v}_k) = \sum_{j_1=1}^{n} \cdots \sum_{j_k=1}^{n} A_{j_1 \cdots j_k} v_{1 j_1} \cdots v_{k j_k}$$

Tensor Product

Well, after looking at that equation, we would naturally feel a bit overwhelmed. Fortunately, we can actually understand it more intuitively.

To understand **K-Linear** more intuitively, we will define a new multiplication rule called **Tensor Product**, which is actually the extension of **Outer Product**.

Given two Tensors, $A \in T^j(\mathbf{V})$ and $B \in T^k(\mathbf{V})$, the Tensor Product between A and B is defined as:

$$A \otimes B(v_1, \cdots, v_j, v_{j+1}, v_{j+k}) = A\left(v_1, \cdots, v_j\right)B(v_{j+1}, v_{j+k})$$

The Tensor Product of $T^j(\mathbf{V})$ and $T^k(\mathbf{V})$ will result in Higher Order Tensor.

$$A \otimes B \in T^{j+k}(\boldsymbol{V})$$

Example of Tensor Product

Suppose we have two vectors **v**, **u** ∈ **V** with the basis {$\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{e}_3$, …, $\mathbf{e}_n$}.

Then, Tensor Product between **v** and **u** is:

$$\begin{aligned} \boldsymbol{v} \otimes \boldsymbol{u} &= (\boldsymbol{e_j} \otimes \boldsymbol{e_i})(\boldsymbol{v}, \boldsymbol{u}) \\ &= (\boldsymbol{e_i}.\boldsymbol{v})(\boldsymbol{e_j}.\boldsymbol{u}) \\ &= v^i u^j \end{aligned}$$

As we can see, $\boldsymbol{v} \otimes \boldsymbol{u} \in T^{n+n}(\boldsymbol{V})$. In Matrix Form, v^iu^j can be represented as:

$$\begin{pmatrix} v^1u^1 & \cdots & v^nu^1 \\ \vdots & \ddots & \vdots \\ v^nu^1 & \cdots & v^nu^n \end{pmatrix}$$

In addition, Tensors are **decomposable** in a sense that Tensor Product is simply the **permutable products** of **Covectors**.

Multi-Linear Map

Intuitively, Tensor Product can be thought of as **Joint Mapping** of **Multiple Linear Spaces**.

From this perspective, we can also see Tensors as **Multi-Linear Maps**, which can be constructed by jointly permuting the different Linear Spaces.

As we could recall from **Outer Product**, **Outer Product Space** defines the **Linear Mapping** between two different Linear Spaces.

$\boldsymbol{f}_j \otimes \boldsymbol{e}_i : \boldsymbol{V} \rightarrow \boldsymbol{U}$, where $\mathbf{f_j} \otimes \mathbf{e_i}$ is the basis of Linear Span $\{\mathbf{f_j} \otimes \mathbf{e_i}\}$

Intuitively, we could conclude that **Tensor Product** also defines the **Multi-Linear Mapping** among different multiple Linear Spaces.

$$\boldsymbol{e}_{i_1} \otimes \cdots \otimes \boldsymbol{e}_{i_k} : \boldsymbol{V}^k \rightarrow R$$

Tensor Representation

Now, we will try to represent Tensors with Tensor Products, as the previous representation is a bit overwhelming.

The k-tuples of $\boldsymbol{v}_k$ in **K-Linear Form** of $T^k(\mathbf{V}) = f(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ was previously defined as:

$$f(\boldsymbol{v}_1, \cdots, \boldsymbol{v}_k) = \sum_{j_1=1}^{n} \cdots \sum_{j_k=1}^{n} A_{j_1 \cdots j_k} v_{1 j_1} \cdots v_{k j_k}$$

However, as we can see,

$$\begin{aligned} \boldsymbol{v}_1 \otimes \cdots \otimes \boldsymbol{v}_k &= \left(\boldsymbol{e}_{i_1} \otimes \cdots \otimes \boldsymbol{e}_{i_k}\right)(\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_k) \\ &= v_{1 i_1} \cdots v_{k i_k} \end{aligned}$$

Tensor Representation

Therefore, the k-tuples of $\boldsymbol{v}_k$ in **K-Linear Form** of $T^k(\mathbf{V}) = f(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ can be rewritten as:

$$f(\boldsymbol{v}_1, \cdots, \boldsymbol{v}_k) = \sum_{j=1}^{n} A_{j_1 \cdots j_k} (\boldsymbol{e}_{j_1} \otimes \cdots \otimes \boldsymbol{e}_{j_k})(\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_k)$$

Thus,

$$A = \sum_{j=1}^{n} A_{j_1 \cdots j_k} (\boldsymbol{e}_{j_1} \otimes \cdots \otimes \boldsymbol{e}_{j_k})$$

A is the **Tensor Representation** of $T^k(\mathbf{V})$, and $f(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ represents the **K-Linear Form** of $T^k(\mathbf{V})$.

$$f(\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_k) = A(\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_k)$$

K-Linear Map

Actually, the **Tensor Representation** A for $T^k(\mathbf{V})$ can be generalized for **K-Linear Map**:

$$A = \sum_{j=1}^{n} A_{j_1 \cdots j_k} (\boldsymbol{e}_{j_1} \otimes \cdots \otimes \boldsymbol{e}_{j_k})$$

For k = 0,

$A = A$, which is obviously a scalar

For k = 1,

$$A = \sum_{j=1}^{n} A_{j_1} \boldsymbol{e}_{j_1}$$, which is obviously a vector

For k = 2,

$$A = \sum_{j=1}^{n} A_{j_1 j_2} (\boldsymbol{e}_{j_1} \otimes \boldsymbol{e}_{j_2})$$, which is obviously a matrix

A Matrix or 2-Linear Map can be either **Linear Map or **Bi-Linear Map** depending on the number of input vectors: A(**v**) or A(**v**, **u**).

K-Linear Map

As we know already **Tensor** usually have both **Covariant Components** and **Contravariant Components**.

Therefore, **Tensor Representation** has both as well.

$$A = \sum_{j=1}^{n} A_{j_1 \cdots j_k} (\boldsymbol{e}_{j_1} \otimes \cdots \otimes \boldsymbol{e}_{j_k})$$

contravariant covariant

Actually, ($\mathbf{e}_{j1} \otimes \ldots \otimes \mathbf{e}_{jk}$) represent the **basis** of the **Tensor**, and thus they are **Covariant Components**, while $A_{j1\ldots jk}$ represents the **Cotravariant Components**.

K-Linear Form

The **K-Linear Form** of $T^k(\mathbf{V}) = f(\mathbf{v}_1,\ldots, \mathbf{v}_k)$ can be defined as:

$$f(\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_k) = A(\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_k)$$

It is generally known as: ***Tensor*** A **acting** on a tuple of **vectors** $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$.

If a **0-Linear** Tensor A(Scalar) acts on a vector **v**, then it is just a **Scalar Product**.

$$A(\boldsymbol{v}) = A\boldsymbol{v}$$

If a **1-Linear** Tensor A (Vector) acts on a vector **v**, then it is the same as **Inner Product**.

$$A(\boldsymbol{v}) = A^i v^i$$

However, if a **2-Linear** Tensor A (Matrix) acts on a vector **v**, or two vectors **v** and **u**, then:

$$A(\boldsymbol{v}) = A^{ij} v^i \boldsymbol{e}_j$$, which is a Linear Map

$$A(\boldsymbol{v}, \boldsymbol{u}) = A^{ij} v^i u^j$$, which is a Bi-Linear Map

Change of Basis in K-Linear Map

I think, we have learned the Change of Basis in **Linear Map** before.

Now, we will try to represent a generic **K-Linear Map** by defining change of basis as well.

As we already know, the change of basis is defined by two metric tensors: **forward** and **backward** metric tensors.

$$\boldsymbol{f}_i = \sum P_{ij}\, \boldsymbol{e}_j$$

$$\boldsymbol{e}_i = \sum Q_{ij}\, \boldsymbol{f}_j$$

Change of Basis in K-Linear Map

Let **V** be a N Dimensional Space with the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \ldots, \mathbf{e}_n\}$ and let $\mathbf{V}^k$ be a set of k-tuples $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \ldots, \mathbf{v}_k)$ where $\mathbf{v}_k \in \mathbf{V}$. Then, we will try change the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \ldots, \mathbf{e}_n\}$ to the new basis $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \ldots, \mathbf{f}_n\}$.

$$A = \sum A_{j_1 \cdots j_k} (\boldsymbol{e}_{j_1} \otimes \cdots \otimes \boldsymbol{e}_{j_k}) \qquad \ldots (1)$$

$$A = \sum A'_{j_1 \cdots j_k} (\boldsymbol{f}_{j_1} \otimes \cdots \otimes \boldsymbol{f}_{j_k}) \qquad \ldots (2)$$

We will try to substitute with Metric Tensors $\boldsymbol{f}_i = \sum P_{ij}\, \boldsymbol{e}_j$ in eq. (2), such that:

$$A = \sum A'_{j_1 \cdots j_k} \left(\sum P_{i_1 j_1}\, \boldsymbol{e}_{i_1} \otimes \cdots \otimes \sum P_{i_k j_k}\, \boldsymbol{e}_{i_k}\right)$$

Change of Basis in K-Linear Map

By simplification:

$$A = \sum P_{i_1 j_1 \cdots i_k j_1} A'_{j_1 \cdots j_k} (\boldsymbol{e}_{i_1} \otimes \cdots \otimes \boldsymbol{e}_{i_k})$$

Likewise, we can show that by substituting $\boldsymbol{e}_i = \sum Q_{ij} \boldsymbol{f}_j$ in eq. (1)

$$A = \sum Q_{i_1 j_1 \cdots i_k j_1} A_{j_1 \cdots j_k} (\boldsymbol{f}_{i_1} \otimes \cdots \otimes \boldsymbol{f}_{i_k})$$

Generic K-Linear Map

Generally, a **K-Linear Map** or **Tensor** A can be defined as:

$$A = \sum P_{i_1 j_1 \cdots i_k j_1} A'_{j_1 \cdots j_k} (\boldsymbol{e}_{i_1} \otimes \cdots \otimes \boldsymbol{e}_{i_k})$$

or

$$A = \sum Q_{i_1 j_1 \cdots i_k j_1} A_{j_1 \cdots j_k} (\boldsymbol{f}_{i_1} \otimes \cdots \otimes \boldsymbol{f}_{i_k})$$

Tensor Notation

Now, we will look into Tensor Notation. Previous, we know that we can represent **K-Linear Form** $f(\mathbf{v}_1,\ldots,\mathbf{v}_k)$ with:

$$f(\boldsymbol{v}_1,\cdots,\boldsymbol{v}_k)=\sum_{j=1}^{n}A_{j_1\cdots j_k}(\boldsymbol{e}_{j_1}\otimes\cdots\otimes\boldsymbol{e}_{j_k})(\boldsymbol{v}_1,\boldsymbol{v}_2,\cdots,\boldsymbol{v}_k)$$

And,

$$\boldsymbol{v}_1\otimes\cdots\otimes\boldsymbol{v}_k=(\boldsymbol{e}_{i_1}\otimes\cdots\otimes\boldsymbol{e}_{i_k})(\boldsymbol{v}_1,\boldsymbol{v}_2,\cdots,\boldsymbol{v}_k)$$

Thus, we can represent $f(\mathbf{v}_1,\ldots,\mathbf{v}_k)$ as:

$$f(\boldsymbol{v}_1,\boldsymbol{v}_2,\cdots,\boldsymbol{v}_k)=\sum A_{j_1\cdots j_k}(\boldsymbol{v}_1\otimes\cdots\otimes\boldsymbol{v}_k)$$

Tensor Notation

If we represent **K-Linear Form** $f(\mathbf{v}_1,\ldots,\mathbf{v}_k)$:

$$f(\boldsymbol{v}_1,\boldsymbol{v}_2,\cdots,\boldsymbol{v}_k)=\sum A_{j_1\cdots j_k}(\boldsymbol{v}_1\otimes\cdots\otimes\boldsymbol{v}_k)$$

Then, in addition to **Vectors** in Vector Spaces ($\mathbf{V}\otimes\mathbf{V}\otimes\ldots\otimes\mathbf{V}$), we can also applies it to **Covectors** in Dual Spaces ($\mathbf{V^*}\otimes\mathbf{V^*}\otimes\ldots\otimes\mathbf{V^*}$):

$$f=\sum A_{j_1\cdots j_k}(\underbrace{\boldsymbol{v}_1\otimes\cdots\otimes\boldsymbol{v}_k}_{\text{Vectors}}\otimes\underbrace{v^*{}_1\otimes\cdots\otimes v^*{}_k}_{\text{Covectors}})$$

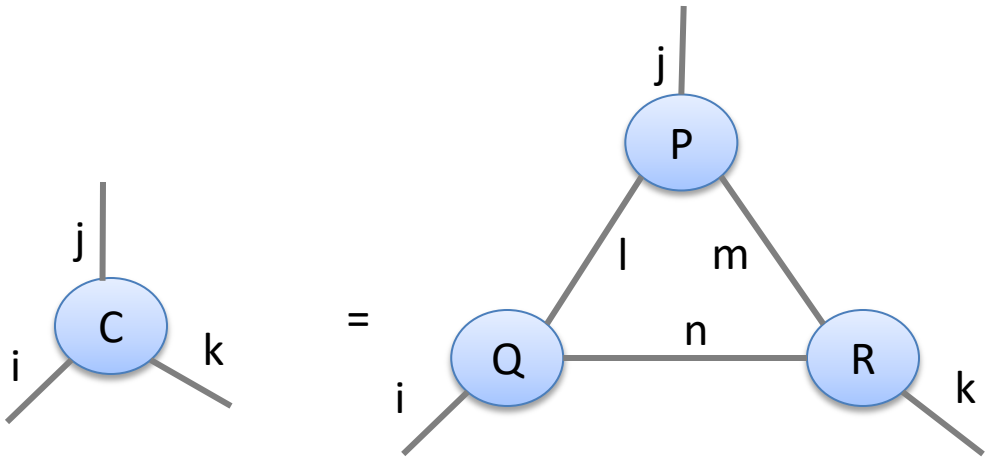
Thus, in **Tensor Notation**, a **Mixed Tensor** can be defined as:

$$T_s^r(\boldsymbol{V}^*,\boldsymbol{V})=T(v^*{}_1\otimes\cdots\otimes v^*{}_r\otimes\boldsymbol{v}_1\otimes\cdots\otimes\boldsymbol{v}_s)$$

$$T_s^r(\boldsymbol{V}^*,\boldsymbol{V})=T_0^r(\boldsymbol{V}^*)\otimes T_s^0(\boldsymbol{V})=T_s^r$$

Tensor Contraction

Tensor Contraction can be defined as **Reducing Tensor Rank and Combing them** by **Summing Over the Indices**, such that:



$$C_{ijk} = \sum_{lmn} P_{ljm} Q_{iln} R_{nmk}$$

Tensor Contraction allows us to simplify complex tensors by reducing their ranks, and combing them.

Tensor Contraction

Previously, we defines **Duality** in such a way that:

$$g^i . \boldsymbol{g}_j = g^i(\boldsymbol{g}_j) = \delta^i_j$$, where g^i is the **dual basis** of $\mathbf{g_j}$.

Also,

$$g^{ij} . \boldsymbol{g}_{jk} = \delta^i_k$$

This rule is particularly important as it can also be used to perform **Tensor Contraction**, which also enable us to simplify Complex Tensor, as it reduces the Tensor Ranks by 2.

$$T^i_j g^j . \boldsymbol{g}_i = T^i_j \delta^j_i = T^i_i \quad \text{or} \quad T^i_{jl} g^{jm} . \boldsymbol{g}_{mn} = T^i_{jk} \delta^j_n = T^i_{kn}$$

Suppose, we have a Complex Tensor: $T^i_{jk} B^j_m$

With **Duality**, we can also simplify it in such a way that: $T^i_{jk} B^j_m = U^i_{km}$

Alternation

In order to understand **Tensor Contraction** more formally, we will have to look into another important aspect of Tensors, known as **Alternation**, defined by **Symmetry** along with **Duality** and **Orthogonality**.

I think we have learned briefly about Symmetry before, along with **Levi-Civita Symbols**.

At any rate, we will look into **Symmetry Group**, along with **Permutation**, to represent **Alternating Tensor** to define **Exterior Product**, also known as **Wedge Product**, and **Interior Product**, which is actually the Contraction.

Decomposable Tensor

A tensor T is defined with $L^k(\mathbf{V}) = l_1(\mathbf{v}) \otimes \dots \otimes l_k(\mathbf{v})$, where $l_k(\mathbf{v}) \in V^*$, then such tensor T is said to be **Decomposable Tensor**.

$$L^k(\boldsymbol{V}) = l_1(\boldsymbol{v}) \otimes \cdots \otimes l_k(\boldsymbol{v}) \quad \text{,where} \quad l_k(v) \in V^*$$

Suppose we have Linear Map $A: \boldsymbol{V} \to \boldsymbol{U}$ and a decomposable tensor $T \in L^k(\mathbf{V})$, pull-back operation is defined such that:

$$A^*: L^k(\boldsymbol{V}) \to L^k(\boldsymbol{U})$$

$$A^*T(v_1, \cdots, v_k) = T(Av_1, \cdots, Av_k)$$

$$A^*L^k(\boldsymbol{V}) = A^*l_1(\boldsymbol{v}) \otimes \cdots \otimes A^*l_k(\boldsymbol{v})$$

Symmetric Group

Suppose Σ_k be k-element set $\Sigma_k := \{1, 2, \ldots, k\}$, then the permutation σ of order k is defined as a bijective function:

$$\sigma: \Sigma_k \to \Sigma_k$$

The permutation function σ is defined with elementary operations called **Transposition**, such that $\sigma = \tau_{i,j}(k)$. For every $1 < i < j \leq k$, let $\tau_{i,j}$ be:

$$\tau_{i,j}(l) := \begin{cases} j, l = i \\ i, l = j \\ l, l \neq i, j \end{cases}$$

Let S_k be the set of all permutations defined by σ. Then, S_k is known as **Symmetric Group** or **Permutation Group** of Order k.

Sign of Permutation

Let $x_1,\ldots,x_k$ be the coordinate functions on $\mathbf{R}^k$. For $\sigma \in S$, we can define:

$$(-1)^{\sigma} = \prod_{i<j} \frac{x_{\sigma(i)} - x_{\sigma(j)}}{x_i - x_j}$$

$(-1)^{\sigma}$ is known as Sign of Permutation, and if σ is the product of **an odd number of transpositions**, $(-1)^{\sigma} = -1$ and if σ is the product of an **even number of transpositions** $(-1)^{\sigma} = +1$.

For example: if we have co-ordinates (x, y, z), then

$$(-1)^{\sigma} := \begin{cases} x, y, z : (-1)^{\sigma} = +1 \\ y, x, z : (-1)^{\sigma} = -1 \\ z, x, y : (-1)^{\sigma} = +1 \end{cases}$$

Symmetry

Suppose **V** is an n-dimensional vector space and $T \in \mathrm{L}^k(\mathbf{V})$ a decomposable k-tensor. Then, for $\sigma \in S_k$, we can define **Transposition** of Tensor *T* as $T^\sigma \in \mathrm{L}^k(\boldsymbol{V})$:

$$T^\sigma(v_1, \cdots, v_k) = T\left(v_{\sigma^{-1}(1)}, \cdots, v_{\sigma^{-1}(k)}\right)$$

If a Tensor is said to be **Symmetric**, if it remains **unchanged** under **Transposition**:

$$T^\sigma = T$$

However, a Tensor is said to be **Anti-Symmetric**, if it **changes its sign** under **Transposition**:

$$T^\sigma = -T$$

A set of **Anti-Symmetric** Tensors, can be generally defined as **Alternating Tensors**.

Alternating Tensor

T^σ is a linear map such that T^σ: $L^k(\mathbf{V}) \longrightarrow L^k(\mathbf{V})$. T is said to be **Alternating K-Tensor**, if

$$T^\sigma = (-1)^\sigma T$$

We can define $A^k(V)$ as **the set of all alternating k-tensors** in $L^k(V)$. In fact, $A^k(V)$ is a vector subspace of $L^k(V)$, which represents all of **Anti-Symmetric Tensors**.

To represent $A^k(V)$, we will define a **Alternation Function** ρ, to define **All Transpositions** on $T \in L^k(V)$, such that:

$$\rho(T) = \sum_{\tau \in S_k} (-1)^\tau T^\tau$$

Since S_k has **k-factorial** number of Transpositions, if $T \in A^k(V)$, then :

$$T = \frac{\rho(T)}{k!}$$

Basis of Alternating Tensor

Suppose $e_1,\ldots,e_n$ be a basis of **V**, then let us define $\boldsymbol{e^*_I}$ as the tensor product of $e^*_i \in V^*$:

$$e_I^* = e_{i_1}^* \otimes \cdots \otimes e_{i_k}^*$$

We can also define ψ_I,

$$\psi_I = \rho(e_I^*)$$

Then,

$$\psi_I(e_{j_1}, \cdots, e_{j_k}) = \begin{cases} 1, I = J \\ 0, I \neq J \end{cases}$$

If $T \in A^k(\boldsymbol{V})$, then T can be define as

$$T(e_{j_1}, \cdots, e_{j_k}) = \sum_I C_I \psi_I (e_{j_1}, \cdots, e_{j_k}) = C_J$$

Thus, ψ_I is the basis of $A^k(\boldsymbol{V})$.

Kernel of Alternation Function

As we can see, the **image** of Alternation Function $\rho: L^k(\boldsymbol{V}) \to L^k(\boldsymbol{V})$ is the Subspace of $L^k(\mathbf{V})$, which is $A^k(\mathbf{V})$. Then, the **kernel** of Alternation Function ρ is defined as $\daleth^k(\mathbf{V})$.

In fact, the **kernel** $\daleth^k(\mathbf{V})$ of Alternation Function is made up of **Repeating Alternating K-Tensors** or **Symmetric Tensors**: for some of ψ_I is repeating if $i_r = i_s$ for some $r \neq s$, which will map to $S^k(\mathbf{V})$, which actually is a *Null Space* of ρ.

Therefore,

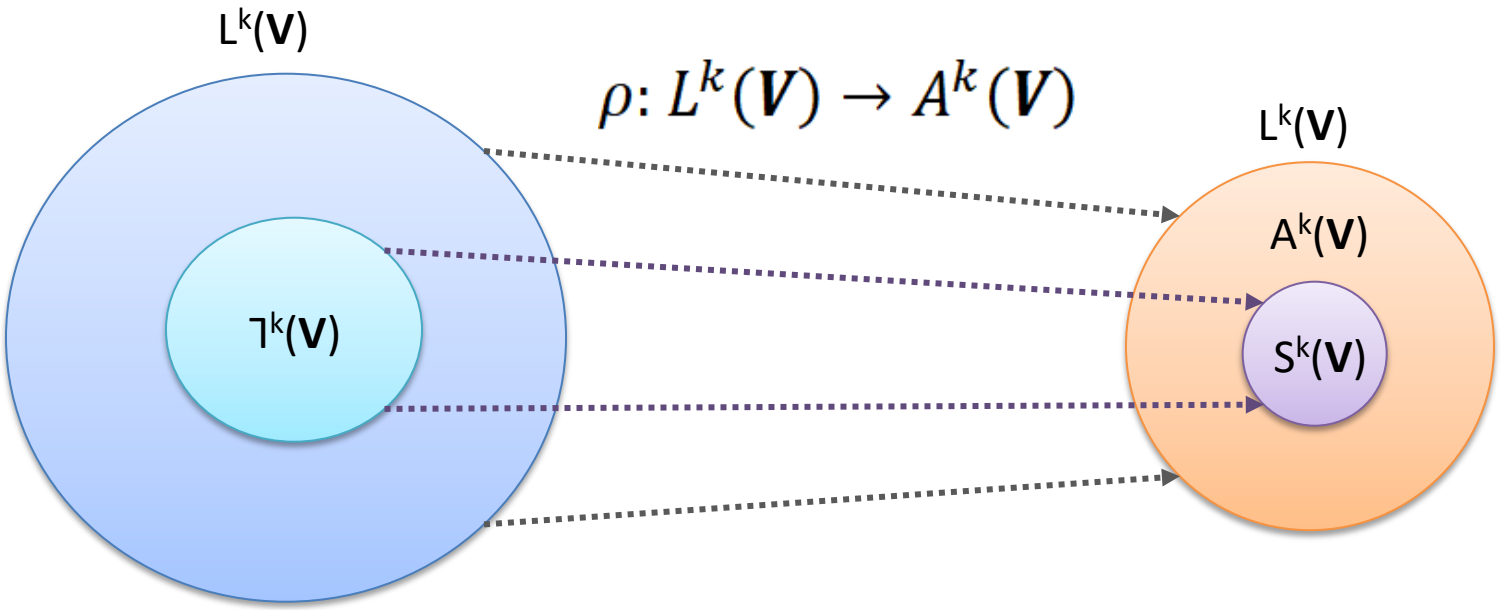
$$\rho(T) = T^\sigma = (-1)^\sigma T + S \text{ , where } (-1)^\sigma T \in A^k(\boldsymbol{V}), S \epsilon S^k(\boldsymbol{V})$$

Thus, every element of $T \in L^k(\boldsymbol{V})$, can be uniquely written by the sum of $A \in A^k(\boldsymbol{V})$ and $S \in S^k(\mathbf{V})$, such that $T = A + S$.

We can actually visualize $A^k(\mathbf{V})$ and $S^k(\mathbf{V})$, along with $\daleth^k(\mathbf{V})$:

$$T^\sigma - (-1)^\sigma T = S$$

Kernel of Alternation Function



$$T^k(\boldsymbol{V}) = kernel(\rho: L^k(\boldsymbol{V}) \rightarrow A^k(\boldsymbol{V}))$$

The Space $\Lambda^k(V^*)$

Then, the Space $\Lambda^k(V^*)$ is defined as the Quotient Space of $L^k(\mathbf{V})$ by $T^k(\mathbf{V})$, such that:

$$\Lambda^k(\boldsymbol{V}^*) = L^k(\boldsymbol{V})/T^k(\boldsymbol{V})$$

Then, we can also define a function π such that:

$$\pi\colon L^k(\boldsymbol{V}) \to \Lambda^k(\boldsymbol{V}^*)$$

As we can see $\Lambda^k(V^*)$ is the Isomorphic Space of $A^k(\mathbf{V})$. We can visualize it as well.

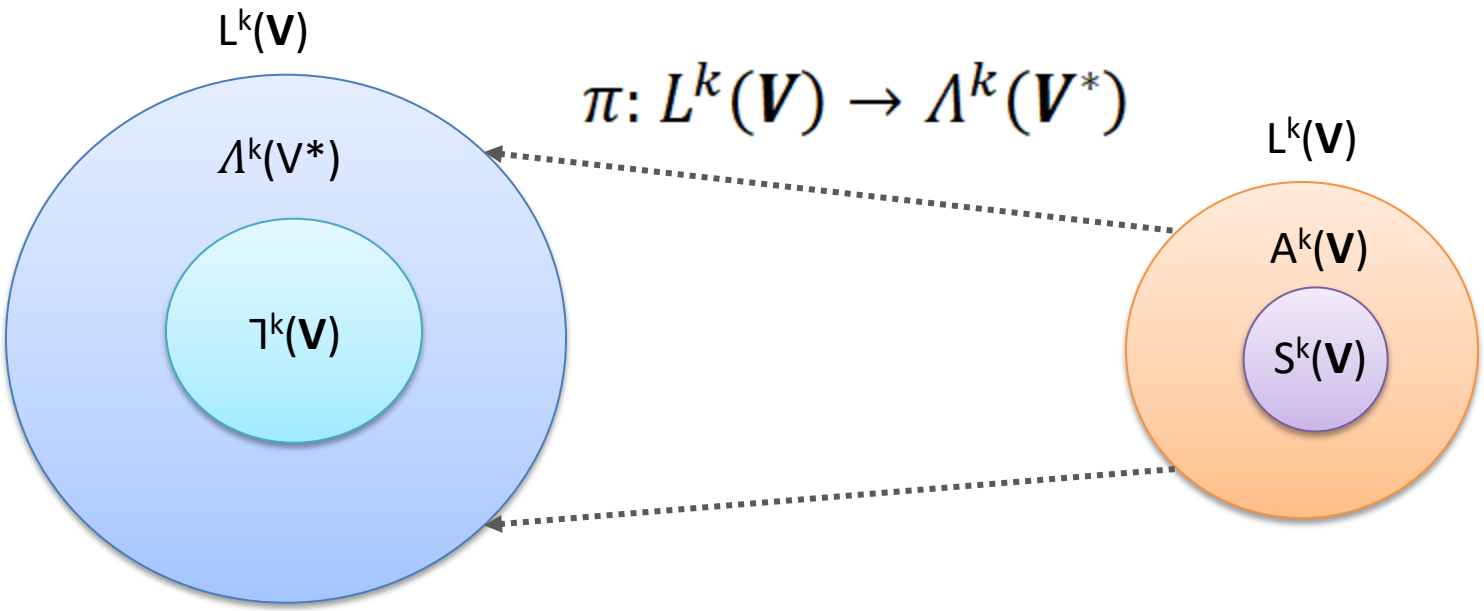
$$\rho\colon L^k(\boldsymbol{V}) \to A^k(\boldsymbol{V})$$

$$\pi\colon L^k(\boldsymbol{V}) \to \Lambda^k(\boldsymbol{V}^*)$$

$$\pi(T) = \frac{\rho(T)}{k!}$$

Actually, the Space $\Lambda^k(V^*)$ defines **Anti-Symmetric** or **Alternating** K-Tensors, which also known as **Exterior K-Forms**.

The Space $\Lambda^k(V^*)$



$$\Lambda^k(\boldsymbol{V}^*) = L^k(\boldsymbol{V})/T^k(\boldsymbol{V})$$

Exterior Product

With the definition of the Space $\Lambda^k(V^*)$, we can now define **Exterior Product**, also known as **Wedge Product**, in the Space $\Lambda^k(V^*)$.

Given $\omega_i \in \Lambda^{ki}(\boldsymbol{V}^*)$, we can find a $T_i \in L^{ki}(\boldsymbol{V})$ with $\omega_i = \pi(T_i)$. If $T_1 \in L^{k1}(\boldsymbol{V})$, $T_2 \in L^{k1}(\boldsymbol{V})$, and $T_1 \otimes T_2 \in L^{k1+k2}(\boldsymbol{V})$, then the wedge product $\omega_1 \wedge \omega_2$ is defined as:

$$\omega_1 \wedge \omega_2 = \pi(T_1 \otimes T_2)$$

$$\omega_1 \wedge \omega_2 = \frac{\rho(T_1 \otimes T_2)}{k_1!\, k_2!}$$

In addition,

$$\omega_1 \wedge \omega_2 = (-1)^{k_1 k_2} \omega_2 \wedge \omega_1$$

Exterior Product

If l_k is **1-Form**, also known as **Exterior 1-Form**, such that $l_1, \ldots, l_k \in V^* = \Lambda^1(V^*)$, and also if $T = l_1 \otimes \cdots \otimes l_k$, then:

$$l_1 \Lambda \cdots \Lambda\, l_k = \pi(l_1 \otimes \cdots \otimes l_k) = \pi(T)\, \epsilon\, \Lambda^k(V^*)$$

However, for any **Exterior K-Form** $\Lambda^k(V^*)$, it can be written in **Determinant Form**:

$$l_1 \Lambda \cdots \Lambda\, l_k(v_1, \cdots, v_k) = \begin{vmatrix} l_1(v_1) & \ldots & l_1(v_k) \\ \vdots & \ddots & \vdots \\ l_k(v_1) & \ldots & l_k(v_k) \end{vmatrix}$$

Exterior K-Form

Previously, we are aware that T ∈ $L^k(V)$ can be written in **Linear K-Form** with **Outer Product**.

$$A = \sum_{j=1}^{n} A_{j_1 \cdots j_k} (l_{j_1} \otimes \cdots \otimes l_{j_k})$$

Likewise, we can write in **Exterior K-Form** if T ∈ $A^k(V)$ or ω ∈ $\Lambda^k(V^*)$:

$$A = \sum_{j=1}^{n} A_{j_1 \cdots j_k} (l_{j_1} \Lambda \cdots \Lambda\, l_{j_k})$$

, where $l_1 \Lambda \cdots \Lambda\, l_k = \pi(l_{j_1} \otimes \cdots \otimes l_{j_k})$

Interior Product

Now, we will define another Product Rule, known as **Interior Product**, which is also the fundamental principle behind **Tensor Contraction**.

As we already know, $\Lambda^k(V^*)$ is defined as the **Quotient Space** of $L^k(\mathbf{V})$ by $\mathcal{I}^k(\mathbf{V})$, which is the $\Lambda^k(V^*)$ Space of **Exterior k-Forms**.

However, we can reduce it to $\Lambda^{k-1}(V^*)$ Space of (k-1)-Forms by **Interior Product**, also known as **Contraction**, of Tensor T with vector **v**.

Interior Product

Let $\boldsymbol{V}$ be a vector space and k a non-negative integer. Given $T \in \mathrm{L}^k(\boldsymbol{V})$ and $\boldsymbol{v} \in \boldsymbol{V}$. Let $\iota_v T$ be the be the $(k-1)$-tensor which takes the value:

$$(\iota_v T)(v_1, \cdots, v_{k-1}) := \sum_{r=1}^{k} (-1)^{r-1}\ T\left(v_1, \cdots, v_{r-1}, v, v_{r+1}, \cdots, v_{k-1,}\right)$$

As we can see, with the r[th] position of **v**, we will sum up all the terms from r = **1** to r = **k**:

$$T(v, v_1, \cdots, v_{k-1})$$

to

$$(-1)^{k-1} T(v_1, \cdots, v_{k-1}, v)$$

Now, if T is the decomposable k-tensor $l_1 \otimes \cdots \otimes l_k$, then

$$\iota_v T = \sum_{r=1}^{k} (-1)^{r-1}\ l_r(v)\left(l_1, \cdots, l_{r-1,}, \hat{l}, l_{r+1}, \cdots, l_{k-1,}\right)$$

Interior Product

If $T_1 \in L^p(\boldsymbol{V})$ and $T_2 \in L^q(\boldsymbol{V})$, then

$$\iota_v(T_1 \otimes T_2) = \iota_v T_1 \otimes T_2 + (-1)^P T_1 \otimes \iota_v T_2$$

If T is the decomposable k-tensor $l_1 \otimes \cdots \otimes l_k$ then

$$\iota_v(l_1 \otimes \cdots \otimes l_k) = \iota_v(T \otimes l) = \iota_v T \otimes l + (-1)^P l \otimes \iota_v T$$

Now, let π be the projection of $L^k(\boldsymbol{V})$ onto $\Lambda^k(V^*)$ and for $\omega = \pi(T) \in \Lambda^k(V^*)$:

$$\iota_v \omega = \pi(\iota_v T)$$

Thus, **Interior Product** ι_v is the Linear Map $\iota_v : \Lambda^k(\boldsymbol{V}^*) \to \Lambda^{k-1}(\boldsymbol{V}^*)$

Operation on Λ^k(V*)

We already know that for Decomposable Tensor T, we can define:

$$A^*T(v_1, \cdots, v_k) = T(Av_1, \cdots, Av_k)$$

Now let ω be an element of Λ^k(V*) and let $\omega = \pi(T)$ where T is in L^k(**V**).

$$A^*\omega = \pi(A^*T)$$

$$A^*(\omega_1 \wedge \omega_2) = A^*(\omega_1) \wedge A^*(\omega_2)$$

Determinant

Let $\boldsymbol{V}$ be a n-dimensional vector space. Then the dimension of $\Lambda^n(V^*)$ is **dim**{ $\Lambda^n(V^*)$} = 1.

As we can see **dim** { $\Lambda^k(V^*)$}: $\binom{n}{k} = \frac{n!}{(n-k)!\,k!}$

Therefore, **dim** { $\Lambda^n(V^*)$} = $\binom{n}{n} = 1$, and thus, $\Lambda^n(V^*)$ is one dimensional vector space. Then,

$$A^*: \Lambda^n(\boldsymbol{V}) \to \Lambda^n(\boldsymbol{V}^*) =$$ Scalar Constant

Thus,

$$A^*\omega = \det(A)\omega$$

Determinant

Let $\boldsymbol{V}$ and **U** be n-dimensional vector spaces and let $\{\boldsymbol{e_1},\ldots,\boldsymbol{e_n}\}$ be a basis and $\{e^*_1,\ldots,e^*_n\}$ dual basis for $\boldsymbol{V}$, and $\{\boldsymbol{f_1},\ldots,\boldsymbol{f_n}\}$ a basis and $\{f^*_1,\ldots,f^*_n\}$ dual basis for **U**.

Then a Linear Map A can be defined as:

$$Ae_j = \sum_i^n a_{ij} f_i$$

Then, for Transpose of Linear Map A* is:

$$A^* f^*_j = \sum_i^n a_{ij} e^*_i$$

Then,

$$A^*\left(f^*_1 \Lambda \cdots \Lambda f^*_n\right) = \sum a_{1k_1} \cdots a_{nk_n} e^*_1 \Lambda \cdots \Lambda e^*_n$$

Determinant

If the multi-index, $k_1,\ldots,k_n$, is repeating, then $e^*_{k1} \wedge \cdots \wedge e^*_{kn}$ is zero, then we can write:

$$k_i = \sigma(i) \qquad i = 1, \cdots, n$$

Then,

$$A^*\left(f^*{}_1 \Lambda \cdots \Lambda f^*{}_n\right) = \sum_{\sigma \in S_n} a_{1\sigma(1)} \cdots a_{n\sigma(n)} \left(e^*{}_1 \Lambda \cdots \Lambda e^*{}_n\right)^\sigma$$

But, $\left(e^*{}_1 \Lambda \cdots \Lambda e^*{}_n\right)^\sigma = (-1)^\sigma e^*{}_1 \Lambda \cdots \Lambda e^*{}_n$

Since $A^*\omega = \det(A)\omega$,

$$A^*\left(f^*{}_1 \Lambda \cdots \Lambda f^*{}_n\right) = \det\left(\left[a_{ij}\right]\right) e^*{}_1 \Lambda \cdots \Lambda e^*{}_n$$

$$\det\left(\left[a_{ij}\right]\right) = (-1)^\sigma \sum_{\sigma \in S_n} a_{1\sigma(1)} \cdots a_{n\sigma(n)}$$

Orientation

Now, we can define the Orientation of a Tensor, as any *Non-Zero* **Exterior K-Form** on **V** can tell us about the orientation of the space **V**.

Let V be an n-dimensional vector space. Then, the **orientation** of **V** is the **orientation** of the one-dimensional **vector space $\Lambda^n(V^*)$**.

If $\{e^*_1,\ldots,e^*_n\}$ is the dual basis, we can orient $\Lambda^n(V^*)$ by requiring that $\{e^*_1 \wedge \cdots \wedge e^*_n\}$ be in the positive component of $\Lambda^n(V^*)$.

Therefore, the ordered basis ($\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n$) of V is positively oriented if $\{e^*_1 \wedge \cdots \wedge e^*_n\}$ is in the positive component of $\Lambda^n(V^*)$.

We can say two bases $\{\mathbf{e_1},\ldots,\mathbf{e_k}\}$ and $\{\mathbf{f_1},\ldots,\mathbf{f_k}\}$ of a vector space **V** define the same orientation of **V** if the **Linear Transformation A** from one to the other has a **Positive Determinant**.